

CS 683: Generative AI

Course Code	CS 683
Course Name	Generative AI
L-T-P-C	3-0-1-4
Prerequisites	CS 671 - Deep Learning and It's Applications
Intended for	Elective for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines, Elective for M.Tech (CSP/Intelligent Systems/CSE) and PhD
Approval	54th BoA (May 2024)

Course Contents

- **Introduction:** Introduction to Generative Models- GenAI, Basics of Deep Learning ,Convolutional Neural Networks, Convolutional Neural Networks -II, Recurrent Neural Networks & LSTM. (4 Hours)
- **Variational Autoencoders:** Deep Dive Into Variational AutoEncoder, Variants of Variational AutoEncoders. (3 Hours)
- **Generative Adversarial Networks:** Theory ofGAN, Variants ofGAN, GAN architectures, Various GAN losses and Convergence analysis ofGAN, Conditional GANS:pix2pix, cycleGAN, Domain adaptation frameworks. (7 Hours)
- **Flow Based Generative Models:** Normalizing Flow Basics, Variational Normalizing Flow and Sampling Basics. (2 Hours)
- **Diffusion Models:** Introduction to Diffusion Models, Deepdive into Diffusion Models -I, Score Based Diffusion Models, Discrete Latent Variable Models, Diffusion Models for Discrete Data. (7 Hours)
- **Hybrid Generative Models:** V AE-GAN, Diffusion guided Style GAN. (2 Hours)
- **Evaluation of Generative Models:** Discussion ofthe scores like FID, Inception Score. (1 Hour)
- **Transformer Based Generative Models:** Introduction to Transformers, Variants of Transformers, BERT Models, GPT Models, Chat GPT, Video and Image generation via. Transformers: VIT, VVIT, CLIP, GLIDE, DALE, Text Generation, Text to Text and Text to Image, Text to Video. (8 Hours)
- **Deep Graph Generative Models:** Graph Neural Networks- Model, Design, Deep Generative Models for Graph Generation and Graph Transformers. (8 Hours)

Laboratory Modules

- Lab to be conducted on a 2-hour slot every week. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class.
- Programming Assignments to Implementing state of art Diffusion Models, GANs,Transformers and Graph Neural Networks.
- Tools- Pytorch or Tensorflow

Text Books

1. Alger fraley, **The Artificial Intelligence and Generative AI Bible:** [5 in 1] The Most Updated and Complete Guide I From Understanding the Basics to Delving into GANs, NLP, Prompts, Deep Learning, and Ethics of AI, 2024
2. David Foster, **Generative Deep Learning**, 2nd Edition, O'Reilly Media, Inc.

References

1. Jakub M. Tomczak, **Deep Generative Modeling**.
2. Keita Broadwater and Namid Stillman, **Graph Neural Networks in Action**.